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20 Years of the Double Pukelsheim System in Switzerland: What Is the Mechanical Effect of the Transition from the Hagenbach-Bischoff Method on the Seat Distribution of Smaller Parties?

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Abstract

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Introduction

On the 18 December 2002 the Swiss Federal Supreme Court (FSC) came to a conclusion, which disrupted the Swiss Electoral System. In BGE 129 | 185 the FSC criticized that in the elections for the Council of the City of Zurich, the thresholds for parties to win a seat varied heavily from district to district. Further did some districts have thresholds for parties to win a seat of up to 34%. Consequently, votes did not have equal effect, which undermines the electoral equality included in Art. 8 (Equality before the law) and Art. 34 (political rights) of the Federal Constitution. Based on this criticism the court decided that the City of Zurich had to implement corrective measures, but could choose their form. (BGE 129 I 185, E. 3, 7.3, 7.6, S. 199–200 ; Schweizerische Eidgenossenschaft, 1999, Arts. 8 and 34) The Government of the Canton of Zurich decided to keep the constituency structure (Regierungsrat des Kantons Zürich, 2014, pp. 3–4). Instead the canton of Zurich commissioned the german mathematician Friedrich Pukelsheim to develop a comprehensive and at the same time minimally invasive solution for the problem. Eventually Pukelsheim came up with a method called "biproportional divisor method with standard rounding" which is today often simply called "double Pukelsheim" or "Biproportional apportionment". (Lüscher, 2024, pp. 9-10)

The distinguishing feature of the Pukelsheimer Proporz, is that it through the so called upper-apportionment artificially creates a single electoral district for the initial allocation of the parliamentary seats to the parties or list groups. Through this mechanism, the difference between seat and voteshares gets minimized and with that also the effective threshold for parties to enter the parliament gets lowered. (Lüscher, 2023, p. 575)

Through its upper-apportionment mechanism and the lowered threshold, the reform is especially relevant for the representation of smaller parties, since votes that would previously have been ineffective in small constituencies can now contribute to a seat gain at the cantonal level. Sandro Lüscher illustrates this mechanical effect with the case of Schwyz in 2016: under the old Hagenbach-Bischoff system, the Green Liberal Party would not have won a single seat, whereas under the new method they obtained three seats (Lüscher, 2025, p. 1-3).

Since Zurich first used the system in 2007, eight further cantons have used it: Schaffhausen (2008), Aargau (2009), Nidwalden (2014), Zug (2014), Schwyz (2016) , Valais (2017) , Uri (2020), Graubünden (2022). In 2025 also the voters from Basel Landschaft decided to implement the Biproportional apportionment from 2027 on (Lüscher , 2024, p.17 ; Kanton Basel-Landschaft, 2025a, p.7 ; Kanton Basel-Landschaft, 2025b) While nine of these cantons already implemented the algorithm, more cantons are currently discussing the implementation too. In Thurgau, the initiative "Mehr Transparenz bei Wahlen im Thurgau" is currently pending at the Grand Council of the Canton and a special parliamentary committee is currently examining the iniatitive, also in the canton of Lucerne, the iniatitive "Faire Wahlen ohne Listenflut" was submitted and the voters will be able to decide on the effectuation (Grosser Rat des Kantons Thurgau, n.d.; Regierungsrat des Kantons Luzern, 2026, pp. 19–20). In the canton of St.Gallen in 2026 an initiative to adapt the algorithm was launched (Initiativkomitee «Damit jede Stimme zählt», n.d.).

The amount of cantons who have already implemented the system or are debating about doing so, suggets that the decision by the FSC had a substantial impact on how Swiss citizens elect their regional parliaments. It seems that the double-Pukelsheimer through its wide implemen-

tation across cantons is a success story. Since the first election under double Pukelsheim in Zurich in 2007, the system has been used in 32 cantonal parliamentary elections.[1]

Taken together, this number of elections and almost 20 years of experience provide a reason to evaluate the impact of the reform, especially on smaller parties. How much has the reform changed smaller parties seat shares and how large is the actual difference in seat allocation between the Hagenbach-Bischoff and Double Pukelsheim system? Are there other major differences, for example in how the reform performs in smaller versus larger parliaments, and how does the smaller parties seat share and gain evolve across successive elections?

The central research gap lies in the fact that the existing literature mainly discusses the Doppelproporz in legal, normative, or system-level terms. It emphasizes improved proportionality, fewer weightless votes, and better opportunities for smaller parties, while also pointing to trade-offs such as territorial distortions and counterintuitive seat allocations (Lüscher, 2023, 2024, 2025). The closest precedents for the calculation used in this thesis are the counterfactual comparisons by Pukelsheim & Schuhmacher, Lüscher, and Leuzinger. Pukelsheim & Schuhmacher use the actual 2011 Zurich election under Doppelproporz combined with hypothetical list alliances to calculate how the same votes would have been translated into seats under the former Hagenbach-Bischoff system, showing that larger parties receive more seats at the expense of several smaller parties under the old rule (Pukelsheim & Schuhmacher, 2011, Section D, pp. 1590–1592, Table 2). Lüscher performs a similar comparison for the 2016 Schwyz election (Lüscher, 2024, p. 13, Table 3). Leuzinger compared the then most recent elections in the seven cantons that had adopted Doppelproporz by 2018 with hypothetical allocations under their former electoral systems (Leuzinger, 2018). Unlike the present study, his counterfactual assumes that the list alliances observed in the final pre-reform election would have persisted. As a result, Leuzinger's counterfactual and the method used in this thesis yield different seat allocations for some elections. By contrast, Lüscher's principal empirical analysis follows Swiss cantons over time and estimates how outcomes such as disproportionality and party-system fragmentation would probably have developed in the reforming cantons had they not adopted Doppelproporz (Lüscher, 2024, pp. 126–143). Compared with Leuzinger's seven-election study, the main comparison in this thesis covers 30 elections in eight cantons through 2026. It applies a consistent definition of smaller parties to measure their seat gains without assuming that pre-reform list alliances persisted.

Against this background, this thesis examines the mechanical effect of the electoral reform on the parliamentary representation of smaller parties. More specifically, it asks and shows statistically how many seats smaller parties gain through the transition from the Hagenbach-Bischoff system to Double Pukelsheim in Swiss cantonal parliamentary elections. It further examines whether the reform performs differently in smaller and larger parliaments and how the seat-gains of smaller parties evolve across sucessive elections. Its contribution is the systematic application of the counterfactual calculation to 32 cantonal elections from which

[1]The 32 elections comprise Zurich (2007, 2011, 2015, 2019, and 2023; Amt für Statistik und Daten Kanton Zürich, 2026); Schaffhausen (2008, 2012, 2016, 2020, and 2024; Kanton Schaffhausen, n.d.; Staatskanzlei Kanton Schaffhausen, 2020, 2024); Aargau (2009, 2012, 2016, 2020, and 2024; Department für Finanzen und Ressourcen des Kantons Aargau, 2009; Regierungsrat des Kantons Aargau, 2013; Statistisches Amt Kanton Aargau, n.d.); Nidwalden (2014, 2018, 2022, and 2026; Staatskanzlei Nidwalden, 2014, 2018, 2022, 2026); Zug (2014, 2018, and 2022; Kanton Zug, 2022a, 2022b, 2022c); Schwyz (2016, 2020, and 2024; Bundesgericht, 2012); Valais (2017, 2021, and 2025; Kanton Wallis, 2016, 2020, 2024; Staat Wallis, 2017); Uri (2020 and 2024; Standeskanzlei Uri, 2024a, 2024b); and Graubünden (2022 and 2026; Kanton Graubünden, 2022, 2026).

30 are in the main transition analysis and two hypothetical Graubünden cases are used as a comparison, delivering a consistent definition of smaller parties and aggregated seat-gain measures.

Theory and Literature Review

Theory

This section traces the use of electoral systems in Switzerland, reviews the arguments for reform and compares Double Pukelsheim with Hagenbach-Bischoff. With that it aims to show why the debate about which electoral system to use is one held often and repeatedly. To understand the continous discussion, it presents the underlying arguments brought upfront to change or keep a system. The theory chapter is built up the following way: It starts with a short explanation of the historical situation of electoral systems in Switzerland, secondly the current situation will be explained, thirdly the transition from Hagenbach-Bischoff to Dobule Pukelsheimer in the corresponding cantons will be examined. Then the Pukelsheimer-Proporz algorithm and it's differences to the Hagenbach-Bischoff Proporz will be discussed more in depth.

Historical Situation

Until the late nineteenth century swiss voters elected the regional parliaments and great councils only under majority voting. In a majority voting system, the candidates or parties who receive the relative or absolute majority receive the seats, therefore this systematically favours the dominant parties. The disadvantages for smaller parties let some of the political minorities demand, to change to a proportional electoral system. In 1891 the canton of Ticino, was the first one to use a PR-System to vote for their Grand Council, till the end of World War I , 7 1/2 more cantons had followed. When the socialist and conservative minorities on their third attempt in 1918 achieved, that the National Council will be elected through a proportional system, it boosted the general adoption of it across Switzerland. Most of the cantons Grand Councils aswell as the National Council were then elected with the Hagenbach-Bischoff proportional system. In the beginning of the 21th century, the cantons of Graubünden and Appenzell Innerrhoden were the only ones to still vote their parliaments with the majority system. (Gilg, 2014, "Mehrheits- und Verhältniswahl" section ; Bundeskanzlei, 2013, p. 2) Nowadays the Kanton Appenzell Innerrhoden is the only remaining canton still voting the Grand Council by majority voting (Bundeskanzlei, 2025, p. 18). It can therefore be said, that proportional systems are the dominating electoral systems for cantonal parliaments in Switzerland.

In the last 20 years the prior dominance of the Hagenbach-Bischoff system has been demolished by the wide implementation of the DP. As of 2025, eleven cantons distribute the parliamentary seats with Hagenbach-Bischoff and ten cantons implemented the DP, two cantons namely Glarus and Basel-Landschaft use the Sainte-Laguë method and two namely Ticino and Vaud use the Hare/Niemeyer method (Bundeskanzlei, 2025, p.18). Of the nine cantons analyzed here, eight previously used Hagenbach-Bischoof for at least the majority of their parliamentary seats, only Graubünden priorly used majority voting throughout. Zug's Neuheim constituency and Uri's smaller municipalities also used Majorz rather than Hagenbach-Bischoff (Bundeskanzlei, 2013, Table 1; Bundesgericht, 2010b, E. 5.1; Standeskanzlei Uri,

2020, p. 1; Statistisches Amt des Kantons Zürich, 2011, p. 286; Regierungsrat des Kantons Schaffhausen, 2007, p. 6; Regierungsrat des Kantons Aargau, 2007, p. 18).

The Adoption of Double Pukelsheim in Swiss Cantons

In the following paragraphs the arguments brought up in each reform canton will be discussed in chronological order. The year in the beginning of each entry, marks the year of first application of the double Pukelsheimer in the corresponding canton.

- **2007:** Zurich became the first canton to use Double Pukelsheim, which it had commissioned Friedrich Pukelsheim to develop. The reform followed a 2002 voting-rights complaint about elections to Zurich's city parliament. The Swiss Federal Supreme Court partially upheld the complaint. It found that the thresholds in some districts with up to 33% were massively above the 10% threshold, which is seen as compatible with the prinicples of proportional representation. Further was it claimed, that thresholds vary to much between districts. The ruling mandated the canton of Zurich to reform the system, subsequently it then implemented the Pukelsheimer Proporz with a 5% district level threshold for parties to enter the parliament. (Lüscher, 2024, pp. 9–10, 33-36) In 2021, Zurich added a 3% cantonwide threshold, first applied in the 2023 election. A change to how that threshold is calculated was adopted in 2025 and will first apply in the 2027 election (Regierungsrat des Kantons Zürich, 2024, pp. 2, 5; Regierungsrat des Kantons Zürich, 2025, p. 1).
- **2008:** Schaffhausen : In a 2004 popular vote, Schaffhausen's electorate approved a reduction of the cantonal parliament from 80 to 60 seats. Shortly after a motion calling to reform the electoral law to better protect smaller parties was submitted. After this the Schaffhausen Government proposed a reform of the electoral system and the implementation of the biproportional apportionment. The biproportional apportionment was also preferred, because of the Supreme Courts decisions in Zurich and Aargau. (Regierungsrat des Kantons Schaffhausen , 2007, pp. 1, 3, 13-14) Subsequently the parliament (2007) and the voters (2008) approved the changes. In 2015, the cantonal parliament added a majority clause, which ensures that the strongest party receives at least one seat in the corresponding district. (Lüscher , 2024 , pp. 42- 43) Further did the canton of Schaffhausen decide to prohibit list alliances (Kanton Schaffhausen, 1904/2019, Art. 2b para. 3).
- **2009:** Aargau's voters in the year 2003 approved a proposal in a popular vote, which wanted to lower the amount of parliament seats from 200 to 140. This decision naturally raised the thresholds in the districts and therefore a second constitutional amendment allowing to merge small districts was proposed and approved by the voters. Though the parliamentarians decided against making use of it. As a consequence of the parliaments decision, the electoral equity wasn't given anymore. The Government of Aargau had to act and proposed to the Grand Council to either merge districts or to change to the Double-Pukelsheim. The Grand Council again decided to not change anything. Shortly after this decision by the Grand Council the FSC issued in a verdict, that there are existing inequalities, because of the varying thresholds and district sizes. The court ordered that this needs to be changed. In September 2007, the Grand Council with a narrow majority then decided to implement the Double-Pukelsheimer. In 2011 a 5% local (district) and a 3% global quorum was added after approval of the Grand Council and the voters (Lüscher, 2024, pp. 38 - 40)
- **2014:** Nidwalden: In the year 2010 members of the Green Party in Nidwalden filed a complaint regarding the seat allocation process for the cantonal elections (Lüscher, 2024,

p. 44). That complaint got rejected initially by the Cantonal Court but later the cantonal decision got overruled by the FSC. The Court rejected the argument that historical practice justified the allocation method. (Lüscher, 2024, p. 44- 45 ;Bundesgericht, 2010a, p.353-364) As a reaction to this decision the parliament of Nidwalden came up with multiple possible reforms. The most popular of the suggested reforms, was the change towards the biproportional system. The parliament and the electorate subsequently approved the implementation of the Double-Pukelsheimer. (Lüscher, 2024, p.45)

- **2014:** Zug: In 2010, representatives of several parties formed a complaint at the FSC regarding the differently sized districts. The Federal Supreme Court agreed on the appeals main point. (Bundesgericht, 2010b, E. 4.5, pp. 378, 385) As a reaction to this decision the cantonal Government presented three different possibilities on how to adapt to the FSC ruling. The responsible comission favoured one option: adopting the Double Pukelsheim. The Parliament adopted the reform and added based on the model of Aargau a 5% local and 3% global threshold. (Lüscher, 2024, p. 47-51) A competing proposal to retain the existing system was struck down by the FSC in 2013 (Bundesgericht, 2013, E. 4, pp. 204–205) Voters approved the biproportional reform on 22 September 2013 (Lüscher, 2024, p. 47-51).
- **2016:** Schwyz : The reform process began with the approval of the new cantonal constitution by the electorate. In that it was foreseen, to keep the current electoral system, which means that each municipality is its own electoral district. (Lüscher, 2024, p. 53) In September 2011, the Cantonal Government prepared the Cantonal Council Election for 2012, against the decisions made there three individual complaints were raised at the FSC. The main complaint was that the thresholds in some of constituencies were too high. The Court agreed on the complaints of the three parties partly and stated that the current electoral system doesn't meet the requirements of a proportional representation system. (Bundesgericht, 2012, Sachverhalt A; E. 5.3 and 6) Later also the Federal Assembly refused to approve the critical sections in the new constitution of Schwyz. As a reaction towards this two different initiatives were formed by the political parties in the canton, one wanted a majority system and the other wanted fairer proportional representation with the willingness to remove guaranteed seats for municipialities. The government reacted with the proposal to inaugurate the DP-System while keeping municipialites as districts and preserving their guaranteed seats. The parliament supported putting the counterproposal together with a 1% global threshold to voters. The voters could then decide on either the majority vote initative or the counterproposal. The counterproposal and with that the implementation of the DP was approved. (Lüscher, 2024, pp. 53-55)
- **2017:** Valais : In the year 2004, a complaint at the FSC was filed regarding the varying thresholds in the electoral districts. The FSC then rejected the complaint and argued that there are historical reasons for the shapes of the electoral districts. (Bundesgericht , 2004, E. 1.1, 2.5 and 3) Six years later an alliance from Upper-Valais proposed the popular initiative "Every Vote Counts" and with that the implementation of the DP with only three new districts. In 2011 the canton then set up a commission to develop proposals. (Lüscher, 2024, pp. 58–59) In 2012 multiple voters again filed a complaint at the Supreme Court it then changed its position and ruled that the existing system violated the Federal Constitution (Bundesgericht, 2014, Sachverhalt, S. 109 and E. 5.2). The parliament then approved a constitutional reform introducing biproportional seat allocation, but it also added a guarantee of 35 seats for German-speaking Upper Valais. Voters narrowly rejected the package and also

the initiative “Every Vote Counts” in 2015. To ensure lawful elections in 2017, parliament unanimously adopted a temporary decree in March 2016 without the seat guarantee. The resulting system retained 14 subdistricts, grouped them into six larger districts, and applied biproportional allocation separately within each larger district. Parliament incorporated these rules into the cantonal electoral law in December 2017. (Lüscher, 2024, pp. 61–63) The Canton Valais also implemented a 8 % local (district-level) threshold (Kanton Wallis, 2004b, Art. 154 Abs. 1).

- **2020:** Uri :The canton of Uri used two different systems simultaneously, the districts with at least three seats used a proportional system and the others the majority system (Bundesgericht, 2016, Sachverhalt, p.93). In the year 2015 the Government of Uri prepared itself for the 2016 election and therefore released the seat allocations of the districts. Based on this resolution, a group of voters filed a complaint at the FSC. The court required from the canton a constitutional reform for the next Landrat election while allowing a mixed electoral system. (Bundesgericht, 2016, Sachverhalt, p.93, 5-5.4, 6.5) Shortly after the Socialdemocrats together with the Green Party proposed to either create a single constituency or create new consitutency that would be updated based on the population trends and in accordance with the Federal principles. The motion was rejected by the parliament. Subsequently, the cantonal government proposed to implement the DP-System. The cantonal council though submitted an initative to the Federal Assembly, the aim was to ensure that cantons remain sovereign in their choice of electoral systems. Four years later the National Council, voted against the Iniatitive. The Government Council then asked for the opinion of the political parties, all of them supported the introduction of the Double-Pukelsheimer for the municipalities already using proportional representation. One issue remained contentious:some parties and institutions were favoring to extend the majority system to districts with three and four seats while the others argued to keep the more than five seats rule. In the end the parliament voted for the extension of the majority system. The electorate approved the reform in May 2019. (Lüscher, 2024, pp. 64-67)
- **2022:** Graubünden: Before the Implementation of the biproportional apportionment the canton still used a majority based electoral system. Though there were multiple attempts to introduce a PR system. A main reason for these attempts to fail, was the fear that the protection of local minorities and the 20 single-seat-constituencies would be weakened. (Lüscher, 2024, p.69) In 2017 then multiple parties and individuals filed a complaint at the Federal Supreme Court after the cantons government had released the seat allocations for the districts in the upcoming election. The court partially upheld the complaint, it especially decided that the seat allocation for the smallest district (Avers) and the six largest districts needed to be revised. Specifically the Court found that the guaranteed seats for Aver’s 160 eligible voters violated equality of voting weight, while majoirtiy voting in the six largest constituencies unjustifiably impaired the equal effect of votes on seat allocation. (Bundesgericht, 2019, Sachverhalt, p. 260, E. 7.3, 8.5, 8.6 and 9) As a response to the FSC decision, the government launched a consultation process after which, it presented nine different models. While the government preferred the DP-System, the municipalities and large parties favoured a mixed electoral model. The willingness of the Socialdemocrats to withdraw a separate initative helped build support for Double-Pukelsheim. The parliament then chose the Double Pukelsheim as a compromise, since it made the seat allocation more proportional while keeping representation for small districts. A 3% global (cantonwide)

threshold was also introduced. In 2021 the electorate voted in favor of implementing the DP-System. (Lüscher, 2024, pp. 71-72)

- **2027:** Basel-Landschaft : Unlike other cantons, in Basel-Landschaft there were no implicit quorums of more than 10%, which means that the law complied with the FSC ruling. In 2015 though a parliamentarian requested the cantonal government to examine the comparative effects of the current system with the Double Pukelsheimer. The Government didn't see any need for a reform, the parliamentarian though kept working on the topic and demanded the establishment of a special comission to review the topic. Although the parliament also rejected this proposal, it instructed its executive board to work with external experts to examine how the electoral system could better reflect party vote shares while preserving elected representatives' local ties. A few years later in 2022, the executive board came to the conclusion, that it recommended the implementation of the DP. Through the support of the parliament, the government was then instructed to draft a revision of the electoral law. (Lüscher, 2024, p.86) In the year 2024, the government then submitted the bill and the Cantonal Council voted in favour of it (Regierungsrat des Kantons Basel-Landschaft, 2024 ; Landrat des Kantons Basel-Landschaft, 2024, p. 1052). The new reform was subject to a mandatory referendum, the canton's voters voted for the implementation in 2025 (Kanton Basel-Landschaft, 2025b). The canton also decided to implement a 3% global and a 5% local threshold and stipulated that each constituency has one guaranteed seat (Kanton Basel-Landschaft, 1981/2025, Art. 39a and Art.49 lit.a). The first election under the new system will be held in 2027 (Kanton Basel-Landschaft, 2026).

To summarize, in 8 of 10 reform cantons, a raised complaint at the FSC led to a later reform. Only the cantons Schaffhausen and Basel-Landschaft reform process took place without a Federal Supreme Court ruling. Nevertheless the canton of Schaffhausen's decision was directly influenced by the FSC's decisions in Aargau and Zurich. Throughout all complaints, the concern was electoral equality. In seven of them excessively high natural thresholds in proportional constituencies were central. While in Graubünden the concerns were mainly raised by the use of of the majority system for its large districts and the allocation of seats among constituencies. This demonstrates the influence of the 2002 Federal Supreme Court decision regarding Zurich's City Parliament.

The Hagenbach-Bischoff Method

To be able to understand later the differences between the HB and the DP Method one has to understand the allocation under the priorly used Hagenbach-Bischoff system.

Under Hagenbach-Bischoff, seats are allocated separately in each constituency. Let M denote the number of seats in the constituency, v_i the votes for an eligible list i, and V the total list votes used to calculate the distribution number. The distribution number D is the smallest integer strictly greater than $V/(M+1)$. In the first allocation, each list receives the integer part of its votes divided by D (Bundeskanzlei, 2025, p. 41):

$$D = \left\lfloor \frac{V}{M+1} \right\rfloor + 1, \quad s_i = \left\lfloor \frac{v_i}{D} \right\rfloor.$$

If the first allocation leaves seats unassigned, the remaining seats are awarded one at a time. For each list, its votes are divided by the number of seats it currently holds plus one. The list

with the highest quotient receives the next seat; its quotient h_i and the allocated seats s_i are then recalculated, and the process continues until all M seats have been filled:

$$h_i = \frac{v_i}{s_i + 1}.$$

This calculation yields the same allocation as the D'Hondt divisor method with downward rounding (Bundeskanzlei, 2025, p. 41). List alliances, eligibility thresholds, and rules for resolving equal quotients depend on the applicable electoral law. The counterfactual in this thesis applies the relevant former cantonal rules within each constituency, as detailed in the Data Cleaning section.

The Double Pukelsheimer Method

Based on the weaknesses of the Hagenbach-Bischoff Method and the FSC decision, the canton of Zurich comissioned the mathematician Friedrich Pukelsheim to develop a new method (Lüscher, 2024, p.10). The following section explores how the algorithm works and how it makes sure that it complies with the Federal Law and mitigates the HB weaknesses.

Based on Article 8 and 34 of the Federal Constitution, there are three principles which have to be followed. These are principle of equal voting value, equality of voting power and equality of electoral value. The equal voting value ensures that each voter within an electoral district gets allocated the identical number of votes. The second principle is the equality of voting power (voting weight guarantee). This requires the population represented per seat to be approximately equal across electoral districts. Thirdly there is the equality of electoral success, which should assure, that all votes contribute equally to the election and that as many votes as possible are taken into account in the allocation of seats. (Pukelsheim & Schuhmacher, 2004, pp. 506-507) The equality of electoral success is violated, when the natural quorum (threshold) in a district is too high, this is given when there are less than 9 mandates within a district. Since then the threshold is higher than the 10% maximum given the FSC jurisdiction and it's calculation method, 1/(M+1) with "M" being the number of mandates. (Pukelsheim & Schuchmacher, 2004, p. 505 ; Bundesgericht, 2005, Regeste) Equality of counting value concerns the formal treatment of votes, while equality of voting power depends on the separately determined ratio of population to seats in each electoral district (Pukelsheim & Schuhmacher, 2004, pp. 507, 515–516). For the traditional application of Hagenbach–Bischoff, the main difficulty lies in equality of electoral success when small districts impose high natural thresholds (Pukelsheim & Schuhmacher, 2004, p. 508; Bundesgericht,2002, E. 7.1.1 and 7.6.2).

To approach this main issue of the Hagenbach-Bischoff, the Double Pukelsheimer or originally called the "Neues Zürcher Zuteilungsverfahren" knows a upper and a lower apportionment (Pukelsheim & Schuhmacher, 2004, p.513).

- **Upper Apportionment** To bypass the natural quorums, in the upper apportionment, the whole canton is seen as one single district. All the seats are then allocated towards the parties. For that the votes from the district-level have to be summarized on a cantonal level, there the first difficulty emerges. Since the district magnitudes vary, the voters also have a different amount of votes, depending on from which canton they are from. Therefore to be able to summarize on a global level, the following is done. The number of party votes per constituency are taken and divided with the respective number of seats. The resulting quotient gets rounded to the next number and indicates the amount of voters per list. The

rounded quotients, can then be summed across all districts and divided by the cantonal divisor to get each parties total seats. The divisor is set by an official authority, in Zurich e.g. its the Department of the Government Councillor which is responsible for the elections. The department needs to assure, that with the set divisor exactly as many seats can be allocated as there are available. The number of seats which result per party, are rounded again to the next number. (Pukelsheim & Schuchmacher, 2011, p.1584-1585 ; Kanton Zürich, 2003, § 10 and 103)

- **Lower Apportionment** This step assigns each party's cantonwide seats from the upper apportionment to its district lists while preserving each district's predetermined seat total. First, the party votes in each district are divided by a district divisor chosen so that the rounded results add up to the district's seats. These initial allocations may not match the parties' cantonwide entitlements: in the 2011 Zurich election, the SVP would have received 55 seats instead of the 54 assigned in the upper apportionment. To correct such differences, every district list of a party is divided by the same additional list group divisor. A divisor above one reduces the party's unrounded quotients and can remove seats while a divisor below one can add seats. Consider a hypothetical SVP list in District Y whose quotient after division by the district divisor is 2.51, giving it three seats after rounding. If increasing the SVP's list group divisor to 1.01 changes no other SVP list's rounded allocation, this quotient falls to $2.51/1.01 \approx 2.49$, so the list receives two seats and the SVP's total seats match the ones from the upper apportionment. This change though may leave District Y one seat short, requiring an adjustment to its district divisor. The party and district divisors are then adjusted in turn until every party receives its cantonwide entitlement and every district retains its fixed number of seats. (Pukelsheim & Schuhmacher, 2011, pp. 1585–1586)
- **Trade-offs and Criticisms

To summarize, the upper apportionment in the Double Pukelsheim system reduces inequalities in electoral success caused by the high natural thresholds of small constituencies under the previous district-by-district Hagenbach-Bischoff system. It does so by pooling parties votes across the canton before seats are assigned to constituencies. This is also the main improvement of DP regarding the three principles in contrast to Hagenbach-Bischoff.

Expected effects of Transition

Literature Review

As already initially explained in the Introduction, this study does not follow Leuzinger's or the approach of Pukelsheim and Schuhmacher of carrying historical or hypothetical list alliances forward into the counterfactual allocation. List alliances are in this thesis seen as strategic and election-specific rather than fixed institutional parameters. Their formation depends on the parties contesting an election, their relative electoral strength, ideological proximity, district-level incentives, and expectations about possible seat gains (SOURCE?). In Zurich, for example, the last Kantonsrat election under Hagenbach–Bischoff took place in 2003. Assuming that the alliances formed in 2003 would still have applied in 2023 despite two decades of changes in the party system would therefore introduce a strong and unverifiable behavioural assumption. Some parties that formed list alliances in Zurich in 2003 no longer existed or contested the 2023 election. Conversely, parties such as the Green Liberal Party, which contested the 2023 election, did not yet exist in 2003. (Statistisches Amt des Kantons

Zürich 2003, table 3; Statistisches Amt des Kantons Zürich 2026; Grünliberale Partei Kanton Zürich n.d.). Therefore the counterfactual preserves joint lists or list groups represented as a single unit in the observed election data. But it does not reconstruct historical alliances or assume new alliances between separately reported units. This provides a transparent counterfactual that holds the observed electoral configuration constant and isolates the mechanical consequences of applying Hagenbach–Bischoff. List alliances pool the votes of connected lists for the first seat-allocation stage and can therefore affect which lists win seats (Pukelsheim & Schuhmacher, 2011, p. 1590).

Lüscher similarly compares the actual allocation in the 2016 Schwyz election with a simplified Hagenbach-Bischoff counterfactual, finding that the dominant SVP and CVP would have benefited under Hagenbach-Bischoff, whereas the left-wing alliance and particularly the GLP benefited from Double Pukelsheimer in that single election (Lüscher, 2024 , p.13).

The decision not to carry historical list alliances forward or to create hypothetical ones does not imply that parties would have contested the election without alliances under Hagenbach–Bischoff. Rather, their counterfactual alliance decisions cannot be reliably observed or reconstructed. Because alliances can mitigate the disadvantages experienced by smaller parties under Hagenbach–Bischoff (Pukelsheim & Schuhmacher, 2011, p. 1590), excluding unobserved alliances may affect the estimated magnitude of the resulting seat changes.

Hypotheses

To be able to answer the research question: 20 Years of Double Pukelsheim in Switzerland: What Is the Mechanical Effect of the Transition from Hagenbach-Bischoff on the Seat Allocation of Smaller Parties?

To answer the research question, this thesis tests one main hypothesis concerning smaller parties' seat shares. A sub-hypothesis compares the gain across cantons, and an exploratory sub-question examines how it varies across successive elections within each canton. The main hypothesis is based on the theoretical considerations and the literature review and focuses on the mechanical effect of the electoral reform on smaller parties.

Hypothesis 1

For each election e, smaller parties are those with less than 10% of the vote in the analyzed area: the whole canton except in Uri, where only the four municipalities using Double Pukelsheim are included. Let S_e^{DP} and S_e^{HB} be their combined seats under Double Pukelsheim and the counterfactual Hagenbach-Bischoff allocation, respectively, and let M_e be the number of seats allocated through Double Pukelsheim and included in the comparison. Their gain in seat share, shown for each election in Figure 2, is

$$\Delta_e = 100\frac{S_e^{DP} - S_e^{HB}}{M_e} \quad \text{percentage points} .$$

The main hypothesis is that, compared with Hagenbach-Bischoff, Double Pukelsheim increases smaller parties' share of the analyzed seats by more than five percentage points on average across elections. The hypothesis uses the unweighted mean, $\bar{\Delta} = E^{-1}\sum_{e=1}^{E}\Delta_e$, so each election counts equally. The analysis also reports a mean weighted by the number of analyzed seats in each election.

H_0 (null hypothesis): The mean gain in smaller parties' seat share under Double Pukelsheim is at most five percentage points: $\bar{\Delta} \leq 5$.

H_1 (main hypothesis): The mean gain in smaller parties' seat share under Double Pukelsheim exceeds five percentage points: $\bar{\Delta} > 5$.

The 10% threshold is based on that the Swiss Federal Supreme Court's reasoning that direct and natural quorums, understood as barriers to representation within an electoral district, should generally not exceed 10% (BGE 131 I 74, E. 5.3–5.5; Bundeskanzlei, 2013, p. 8).

Sub-hypothesis H1a: Differences between cantons

The gain in smaller parties' seat share may vary across cantons. As an exploratory comparison, cantons where more than 100 seats are allocated through Double Pukelsheim (ZH, AG, and VS) are compared with cantons where at most 100 seats are allocated through Double Pukelsheim (UR, NW, ZG, SH, and SZ).[2] Graubünden is shown separately because its pre-reform system was Majorz and its HB allocation is hypothetical (Bundeskanzlei, 2013, Table 1).

H_0a: The two groups have the same mean canton-level gain in smaller parties' seat share under Double Pukelsheim.

H_1a: The mean canton-level gain in smaller parties' seat share under Double Pukelsheim differs between the two groups.

For each canton, the election-level gains Δ_e are averaged first; the group mean then averages these canton means so that cantons with more observed elections receive no extra weight. The difference is described in percentage points without a directional prediction or a claim that canton size causes it. Former district magnitudes, legal thresholds, party competition, and canton-specific allocation rules could also shape the result.

Exploratory sub-question: Change across successive elections

How does the gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach-Bischoff allocation vary across successive elections within each canton? The analysis follows the seat shares under both allocation rules and their difference Δ_e from each canton's first election under Double Pukelsheim onward. No directional trend is hypothesized, because vote shares, the composition of parties below the 10% threshold, district seat magnitudes, and electoral rules can change between elections.

Analysis

H1: The analysis takes the actual seat allocation under the Pukelsheimer Proporz as its starting point. Based on the available election data at the municipal or district level, it then reconstructs how many seats each party would have obtained under the former Hagenbach-Bischoff system. The comparison between the actual seat distribution under the new system and the hypothetical distribution under the old system makes it possible to identify how many seats each party gained or lost as a result of the reform. In this way, the study assesses the mechanical effect of the electoral-system change on party representation, with a particular focus on smaller parties.

[2] Official seat totals: ZH (Amt für Statistik und Daten Kanton Zürich, 2026); AG (Statistisches Amt Kanton Aargau, n.d.); VS (Kanton Wallis, 2024); UR (Standeskanzlei Uri, 2024b); NW (Staatskanzlei Nidwalden, 2026); ZG (Kanton Zug, 2022c); SH (Staatskanzlei Kanton Schaffhausen, 2024); SZ (Kanton Schwyz, 2024); GR, shown for context (Kanton Graubünden, 2026).

For the purposes of this thesis, a smaller party is defined as an electoral list or list group with a vote share below 10% in the analyzed area. The analyzed area is equal to the whole canton except in Uri, where the vote shares cover only the four municipalities using Double Pukelsheim. Youth lists and other special-interest lists, such as women's lists or pensioners' lists, are treated as separate analytical units whenever they are reported separately in the official election results; their votes and seats are not attributed to a parent party. Accordingly, the term *party* in the analysis refers to the standardized electoral list or list group recorded in the dataset and does not necessarily correspond to a broader organizational party family. The choice of 10% as the threshold is grounded in both legal and normative considerations. In its decision BGE 131 I 74, the Swiss Federal Supreme Court argued that direct and natural quorums, understood as barriers to representation within an electoral district, should generally not exceed 10% (E. 5.3–5.5; see also Bundeskanzlei, 2013, p. 8). Using vote share rather than absolute party size is particularly appropriate because a party may be electorally small in one region while being comparatively strong in another. Low thresholds are closely linked to the principle of Erfolgswertgleichheit which is the equality in the effect of votes on seat allocation, which requires that votes should be translated into seats as equally and effectively as possible (BGE 131 I 74, E. 3.1; Bundeskanzlei, 2013, pp. 8–9). This threshold is especially relevant to the cantons analyzed here: before their reforms, each had at least one district with an electoral threshold of 10% or higher (Lüscher, 2024, p. 115).

Data Cleaning

(STILL NEED TO CLARIFY ON HOW MUCH OF DATA CLEANING IS EXPLAINED HERE AND HOW MUCH ONLY IN APPENDIX)

To be able to analyze the data from the different elections and compare the outcomes. each dataset has to be cleaned. The data For some cantons e.g. Valais and Schwyz the data is very scattered, therefore for there multiple datasets per election have to be cleaned and merged.

The procedure for the data cleaning was the following: For each canton in alphabetical order, for each election the data was cleaned.

To ensure the compatibility of the different datasets, the column names were standardized. The column names were renamed for each dataset to the following: "total_votes", "svp_votes", "sp_votes", "cvp_votes", "fdp_votes", "glp_votes", "bdp_votes", "green_votes", "other_votes". It might be that some of the datasets do not contain all of these columns, since the party landscapes varies geographically and over time.

For the counterfactual, the observed district-level party votes and district seat magnitudes are retained. Where Hagenbach-Bischoff was used before the reform, its allocation is reconstructed under the relevant former cantonal rules, including the former Zurich Wahlgesetz (§§ 87–88b), the pre-reform Aargau legislation, the former Uri Proporzgesetz (Arts. 22–23), and the former Valais law (Arts. 154–156); the canton-specific scripts document the remaining cases (Kanton Zürich, 1983/2005; Kanton Aargau, 2004; Kanton Uri, 2018; Kanton Wallis, 2004a). The HB allocations in Graubünden and in Zug's Neuheim constituency are hypothetical comparisons rather than reconstructions of their former Majorz rules. The Hagenbach–Bischoff counterfactual allocates seats to the standardized electoral units in each election's data. Joint lists reported as one unit remain pooled but no additional alliances are assumed between separately reported units. The analysis does not carry alliances from the last pre-reform election forward, unlike Leuzinger (2018, p. 42). Vote shares for the analyzed

area are derived from the available official *Wählerzahl* or reported vote-share figures; where necessary, the cleaning scripts reconstruct voter numbers from district-level party votes and seat magnitudes. The calculation and rounding procedures vary by canton and source file.

Schaffhausen reduced its parliament from 80 to 60 seats for the first Double Pukelsheim election. The counterfactual holds the later election's 60-seat district structure fixed, so it isolates the allocation method under that structure rather than reproducing every feature of the 2004 election (Regierungsrat des Kantons Schaffhausen, 2007, pp. 1, 6).

The official Double Pukelsheim results (`seats_dp`) are sourced from official cantonal authorities and are documented in the Bibliography. A single lookup table consolidates all official results for easy merging with the Hagenbach-Bischoff counterfactuals. All data processing is implemented in canton-specific R scripts (R/canton_*.R) which apply shared functions for algorithm execution and validation.

Results

Smaller-party representation across elections

ℹ Scope of the canton comparisons

Graubünden. The canton used Majorz throughout before adopting Double Pukelsheim. Its HB allocation is hypothetical, so the observations are shown in grey and excluded from the main transition estimate (Bundeskanzlei, 2013, Table 1, p. 14).

Zug. Neuheim used Majorz before the reform, while the other ten municipalities used proportional allocation. Neuheim participates in the observed canton-wide Double Pukelsheim allocation, so all 80 seats are retained. The HB allocation for Neuheim is a hypothetical uniform-HB comparison, not a reconstruction of its former Majorz rule (Bundesgericht, 2010b, E. 5.1; Kanton Zug, 2022a, 2022b, 2022c).

Uri. Only Altdorf, Bürglen, Erstfeld, and Schattdorf used Double Pukelsheim in the elections analyzed here. They account for 37 of the Landrat's 64 seats; the remaining municipalities used Majorz and are excluded from both allocations in this comparison (Standeskanzlei Uri, 2020, p. 1). Uri's vote shares, including the less-than-10% classification of smaller parties, are based only on these four municipalities, not on the whole canton.

Valais. Upper apportionment takes place separately within six electoral regions. If for example, the SVP/UDC contested all six regions, it would enter six independent regional allocations; votes in one region cannot increase its seat entitlement in another. This analysis sums the resulting regional seats under one canton-wide party label and aggregates voter numbers to calculate canton-wide vote shares. The single party row and the classification of parties below 10% are therefore measures for comparison, not the regional units used to allocate seats (Lüscher, 2024, p. 57).

Figure 1 compares the observed Double Pukelsheim allocation with the counterfactual Hagenbach-Bischoff allocation for every cleaned election. For each election, the figure aggregates the seats and vote shares in the analyzed area of all parties below the 10% threshold defined above. The grey line connects the two allocations derived from the same election.

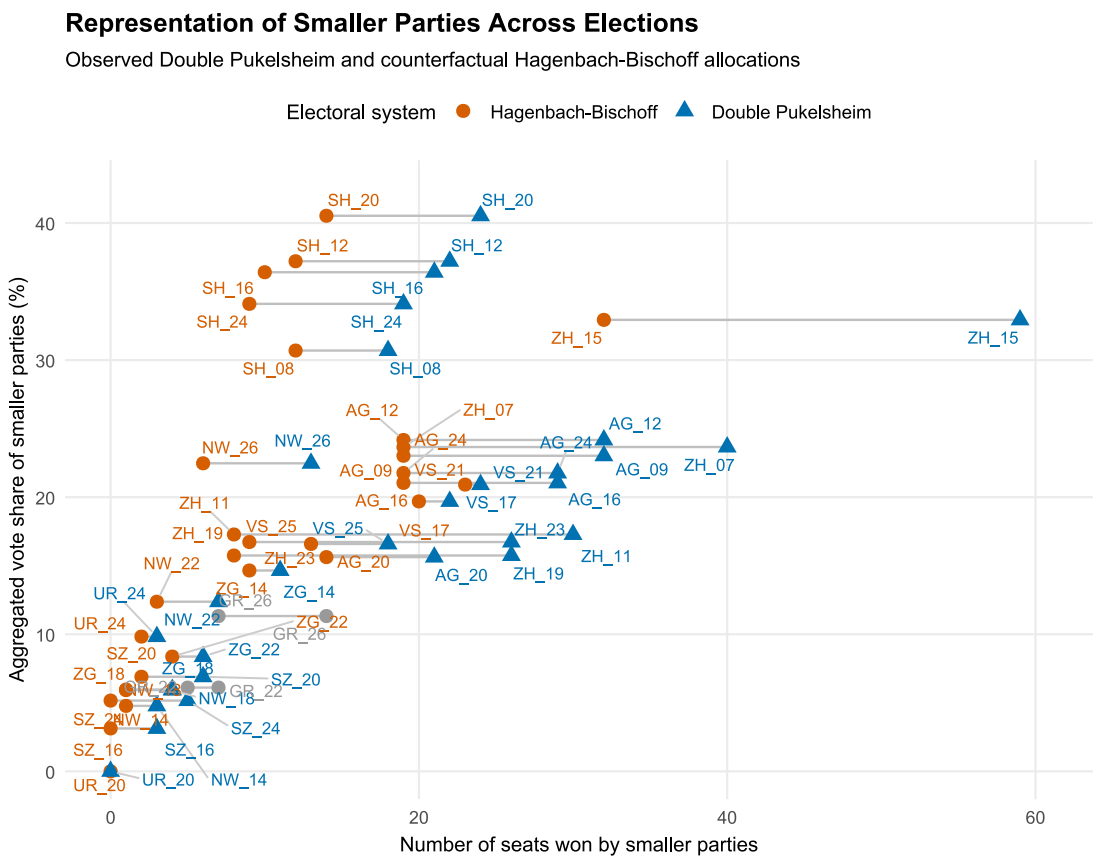


Figure 1: Seats won by smaller parties under Double Pukelsheim and the counterfactual Hagenbach-Bischoff allocation. Smaller parties have less than 10% of the vote share in the analyzed area.

Relative smaller-party representation across elections

Figure 2 shows the same elections using the percentage of analyzed Double Pukelsheim seats won by smaller parties on the x-axis. Each label reports the gain under Double Pukelsheim as a change in seat share, expressed in percentage points.

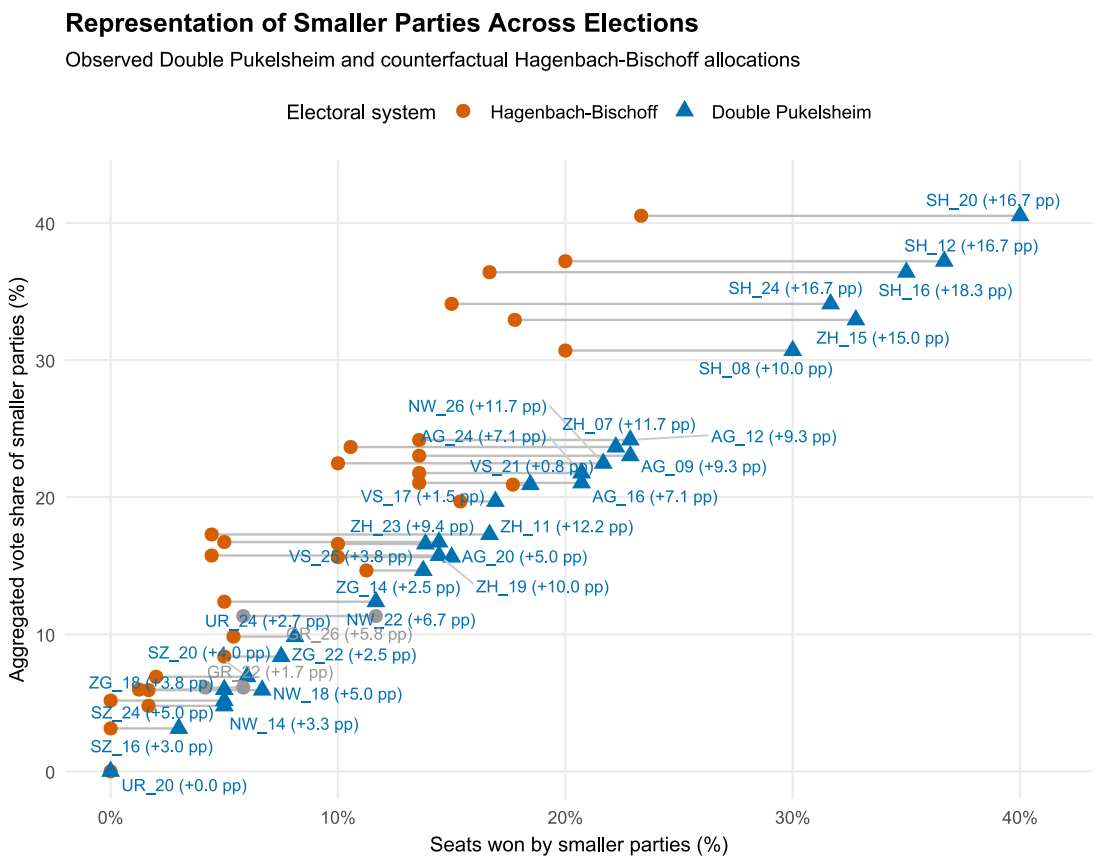


Figure 2: Percentage of analyzed seats won by smaller parties under Double Pukelsheim and counterfactual Hagenbach-Bischoff.

> **i** Hypothesis 1 summary
>
> Across all 30 elections (Graubünden excluded), the unweighted mean gain in smaller parties' share of analyzed seats was 7.7 percentage points (Q1: 3.4; median: 6.9; Q3: 11.2). The mean weighted by each election's number of analyzed seats was 7.9 percentage points. 16 individual elections exceeded five percentage points.

H1a: Differences by Double Pukelsheim seat count

The comparison below uses the same election-level percentage-point gain as H1. It averages elections within each canton before comparing the two groups defined by the number of seats allocated through Double Pukelsheim. Graubünden is only shown for context and excluded from the group means for the same reason as in the main H1 estimate.

Table 1: Mean gain in smaller parties' share of analyzed seats by canton (percentage points). Source: author's aggregation of the official cantonal election datasets.

Canton	Group	Elections	DP seats (parliament)	Mean gain (pp)
ZH	More than 100 DP seats	5	180 (180)	11.7
AG	More than 100 DP seats	5	140 (140)	7.6
VS	More than 100 DP seats	3	130 (130)	2.1
GR	Context only (former Majorz)	2	120 (120)	3.8
SZ	At most 100 DP seats	3	100 (100)	4.0
ZG	At most 100 DP seats	3	80 (80)	2.9
NW	At most 100 DP seats	4	60 (60)	6.7
SH	At most 100 DP seats	5	60 (60)	15.7
UR	At most 100 DP seats	2	37 (64)	1.4

The mean of the canton-level gains is 7.1 percentage points in the group with more than 100 Double Pukelsheim seats and 6.1 percentage points in the group with at most 100. Their difference (more than 100 minus at most 100) is +1.0 percentage points. With only 8 cantons and distinct electoral rules, this is a descriptive comparison rather than evidence of a causal effect of the number of Double Pukelsheim seats.

Interpretation of the Results

Discussion

Conclusion

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Appendix

Appendix 1: Explanation Data Cleaning Process

Appendix 2: Code for Data Cleaning

Declaration of AI-Assistance